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Large language models: an overview of foundational architectures, recent trends, and a new taxonomy

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Abstract
Since the introduction of foundational models such as Bidirectional Encoder Representations from Transformers (BERT) and Generative Pre-trained Transformers (GPT), there has been rapid evolution in both the scale and application of large language models (LLMs). This review provides a concise overview of LLMs, their architecture, training methodologies, and recent innovative applications, focusing on notable models such as the GPT series, BERT, Pathways Language Model (PaLM), and Large Language Model Meta AI (LLaMA), and recently the DeepSeek-R1 model. Additionally, this paper presents a taxonomy for categorizing LLMs based on three critical dimensions: scalability, application domains, and ethical considerations. This taxonomy aims to enable researchers and practitioners to better understand these models in terms of their potential and limitations. Lastly, by reviewing contributions from numerous publications, this study identifies emerging trends, gaps, and opportunities in LLM research, providing a structured guide for future investigations.

Keywords AI, BERT, Deep learning, GPT, LLMs, Transformers, Machine learning

1 Introduction

Large language models (LLMs) have transformed natural language processing (NLP), achieving state-of-the-art performance in tasks such as machine translation, summarization and answering questions [1–4]. LLMs can capture complex linguistic patterns and provide generalizable solutions in a wide range of applications [5–7]. Initially focused on domain-specific tasks, LLMs have evolved through increasingly sophisticated architectures and training methods, enabling broader, more refined language understanding and interaction that extend beyond traditional NLP applications [8–10]. This shift from task-specific models to versatile architectures, as shown by models such as GPT-4, BERT, and PaLM, represents a significant advancement in AI research. Figure 1 shows this evolution of LLMs.

Despite their promise, LLMs face substantial challenges, including the high computational demands for training and deployment, concerns about interpretability, and

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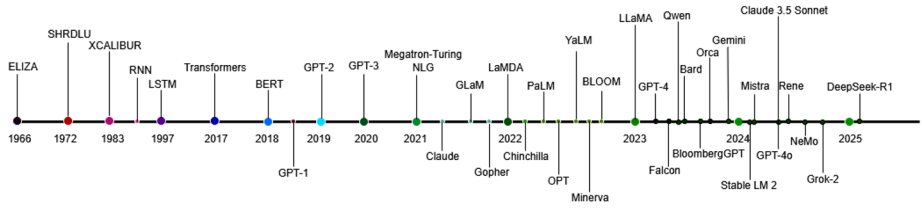

Fig. 1 Evolution of LLMs

issues such as bias and transparency [11–14]. Addressing these limitations is crucial for broader adoption and responsible deployment across sectors. Several recent surveys [15–17] examine LLM applications, primarily in fields such as education and healthcare. For example, Kasneci et al. [15] reviewed the roles of LLMs in education, such as automated feedback and question generation.

Furthermore, Yan et al. [16] discussed ethical and practical challenges in educational contexts. Meanwhile, Abd-Alrazaq et al. [17] address LLMs in medical education, but overlook their broader implications in critical healthcare applications, such as diagnostics and treatment recommendations.

These studies highlight traditional applications but leave significant gaps in the potential of LLMs in newer fields, ethical and social considerations, and challenges around scalability and computational costs. Therefore, this paper aims to address these gaps by offering a holistic review of LLM advancements, applications, and the ethical challenges surrounding their deployment across various industries. The contributions of this paper include:

- The study presents a structured overview of the technological evolution of LLMs, synthesizing developments in architecture, training techniques, and industry-specific applications.
- We propose a conceptual taxonomy, which categorizes LLMs according to scalability, application domains, and ethical considerations. This framework offers a structured method for researchers and practitioners to assess LLMs based on both technical and ethical dimensions.
- The study identifies emerging challenges, including issues of bias, transparency, and energy consumption, emphasizing the need for future research that addresses these limitations to create more accessible and ethical LLM solutions.

The remainder of this paper is structured as follows: Sect. 2 discusses related works, while Sect. 3 presents an overview of LLMs and explores the foundational architectures, and Sect. 4 reviews key training techniques. Section 5 focuses on the evaluation metrics for LLM performance, and Sect. 6 addresses responsible AI and governance frameworks. Section 7 presents emerging LLM applications. Section 8 introduces a conceptual framework for LLM categorization. Section 9 presents challenges and outlines future research directions, and Sect. 10 concludes the study.

1.1 Research methodology

This paper adopts a narrative review approach to examine recent advancements, applications, and challenges in LLMs. A systematic literature search was conducted across primary databases such as IEEE Xplore, ACM Digital Library, SpringerLink, ScienceDirect, and arXiv, focusing on studies published within the last three years. Search terms,

such as "large language models," "transformer architectures," and "LLM challenges," were used to identify key works on foundational architectures, training methodologies, and domain-specific applications, with an emphasis on peer-reviewed journal and conference articles.

The selected studies were thematically organized to address LLM architectures, training processes, ethical considerations, and industry applications. This thematic organization informed the development of a conceptual framework that categorizes LLMs based on scalability, applications, and ethical dimensions. The insights from this review provide a structured perspective on current trends, limitations, and future research opportunities in the evolving field of LLMs.

2 Related works

The rapid advancement of LLMs has led to several research works aimed at understanding their capabilities, limitations, and broader societal impact. For example, Kasneci et al. [15] and Yan et al. [16] provided comprehensive reviews of LLM applications in education, highlighting both the potential and ethical challenges of their deployment in learning environments. These studies focus on tasks such as automated feedback generation and question answering.

In healthcare, Abd-Alrazaq et al. [17] explored the use of LLMs for medical education, specifically focusing on their potential to enhance teaching and learning processes. Similarly, Thirunavukarasu et al. [18] examined the growing role of LLMs in clinical decision-making and medical diagnostics. Furthermore, in the legal field, Williams [19] assessed the capability of large language models (LLMs), such as GPT-3 in automating tasks such as legal document drafting, contract analysis, and case law prediction. Although LLMs show promise in streamlining legal workflows, concerns regarding interpretability, transparency, and bias remain inadequately addressed.

Another promising application of LLMs is within the field of financial technology (fintech). Recent studies have demonstrated the utility of LLMs in tasks such as fraud detection, automated customer service, and financial forecasting. For instance, Papasavva et al. [20] explored the use of LLMs to detect fraudulent transactions by analyzing unstructured textual data from customer interactions. Their study showed that LLMs could identify patterns that traditional machine learning models often miss, thus improving the overall accuracy of fraud detection systems.

In addition to these domain-specific applications, LLMs have been explored for enhancing customer support across various industries. Recent work by Nasseri et al. [21] investigated how LLMs could be integrated into customer service platforms to handle a wide range of queries, from troubleshooting technical issues to providing product recommendations. Their findings suggest that LLMs can effectively reduce response times and improve customer satisfaction by delivering more personalized interactions. However, the study also noted challenges related to context retention, where the models occasionally produced responses that were inconsistent with earlier parts of the conversation, highlighting the need for improved context management techniques in LLMs.

Meanwhile, there has been growing attention on the role of LLMs in creative domains. Tools, such as OpenAI's GPT-3 have been used to generate creative content, including poetry, music lyrics, and screenplays, thereby broadening the scope of what these models can achieve. Bhat et al. [22] conducted a study on how LLMs could assist writers by

generating ideas or drafting initial content. The study found that, while LLMs could provide a valuable starting point for creative projects, their outputs often lacked the detailed understanding of tone and style that human creators bring, necessitating human oversight and refinement. This has sparked a discussion around the ethical considerations of using AI in creative fields, mainly concerning authorship and originality.

While existing surveys have extensively covered traditional applications in education, healthcare, law, and fintech, they generally overlook emerging domains such as cybersecurity, creative content generation, and real-time programming assistance. Additionally, comprehensive discussions about scalability, ethical implications, computational efficiency, bias mitigation, and interpretability are notably limited or absent in current literature, leaving significant research gaps.

This study addresses these gaps by reviewing traditional and novel applications of LLMs, particularly focusing on underexplored areas such as cybersecurity, legal analysis, and creative domains. It also critically evaluates ethical and governance challenges, including bias, transparency, and societal impacts. Furthermore, this work introduces a conceptual taxonomy categorizing LLMs based on scalability, application domains, and ethical impact, offering clear criteria and justifications for model categorization that current literature lacks. By providing structured insights into these overlooked areas, this paper enhances understanding of both technical capabilities and broader implications of LLMs, highlighting avenues for future research and policy development.

3 Foundational architectures

The concept of training large models on vast amounts of text data was first explored in the early 2010s. However, it was not until the introduction of the Transformer architecture in 2017 by Vaswani et al. [23] that LLMs became feasible and effective on a large scale. The Transformer model, with its self-attention mechanism, allowed for much more efficient parallelization during training compared to recurrent neural networks (RNNs) such as long short-term memory (LSTM) networks, which previously dominated NLP tasks [24]. This marked a shift from sequential to non-sequential processing of input data.

Furthermore, LLMs are trained on massive datasets, often containing hundreds of gigabytes of text from sources such as books, websites, and articles. Through the use of patterns, grammar, and even world knowledge, LLMs can perform a variety of tasks, as illustrated in Fig. 2. The following subsections provide a brief overview of foundational LLM architectures.

3.1 GPT

The Generative Pre-trained Transformer (GPT), introduced by Radford et al. [25], is a foundational architecture in the evolution of LLMs. Built on the Transformer framework (Fig. 3), GPT applies self-attention to capture long-range dependencies in text, offering improvements over earlier sequential models such as RNNs. Unlike bidirectional models like BERT, GPT employs an autoregressive approach, predicting each token based on preceding tokens. This design makes it effective for tasks such as text generation and summarization [26, 27].

GPT leverages only the decoder component of the Transformer architecture, optimized for generative tasks. A key innovation is the multi-head attention mechanism,

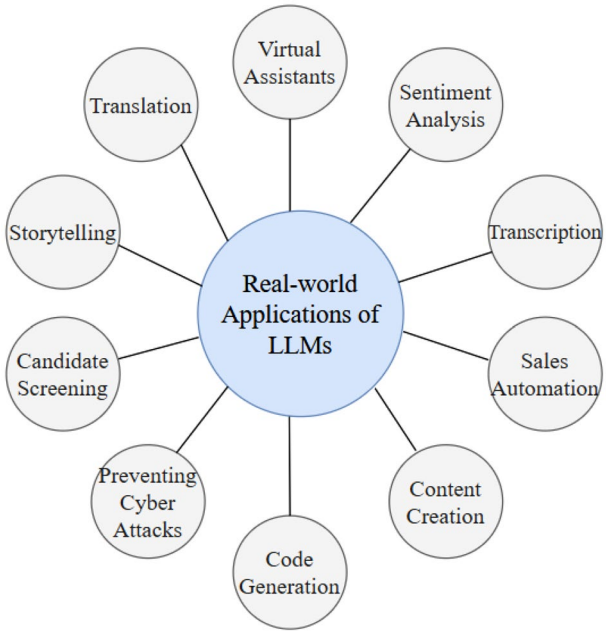

Fig. 2 Applications of LLMs

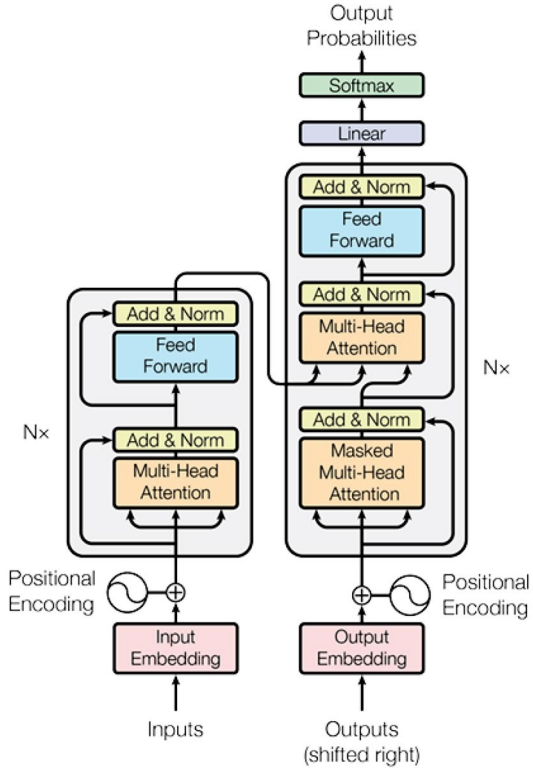

Fig. 3 Transformer architecture [23]

which enables the model to simultaneously consider multiple positions in the input sequence. The attention is computed as:

$$\text{Attention}\,(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V, \tag{1}$$

where Q, K, and V denote the query, key, and value matrices, respectively, and d_k is the dimension of the key vectors. Scaling has been central to GPT's evolution. GPT-2 (1.5 billion parameters) introduced stronger generalization, while GPT-3 (175 billion) demonstrated high-quality few-shot learning and translation [8]. GPT-4 extended the architecture with multimodal capabilities, supporting both text and image inputs [28]. Variants like GPT-4o and GPT-o1 mini further balance performance with computational efficiency, broadening accessibility.

3.2 BERT

BERT, developed by Devlin et al. [29], introduced a bidirectional approach to contextual understanding by analyzing both preceding and succeeding tokens in a sequence. This design enables the model to capture intricate dependencies between words, improving performance in tasks such as sentiment analysis, question answering, and named entity recognition. BERT's pre-training involves two core objectives: masked language modeling (MLM) and next sentence prediction (NSP). In MLM, certain tokens are masked and predicted using the surrounding context:

$$\mathcal{L}_{MLM} = -\sum_{i=1}^{n} \log P(x_i | x_{\setminus i}, \theta), \tag{2}$$

where x_i denotes a masked token and $x_{\setminus i}$ represents unmasked tokens. Meanwhile, NSP determines whether a sentence logically follows another, enhancing inter-sentence coherence. These mechanisms are illustrated in Fig. 4. BERT's architecture has inspired several optimized derivatives. RoBERTa [30] removes NSP and trains on larger datasets, while ALBERT [31] reduces model size via parameter sharing. BigBird [32] introduces sparse attention to handle longer sequences efficiently, and DeBERTa [33] improves contextual encoding with disentangled attention and an enhanced decoder. These variants extend BERT's capabilities to a broader set of NLP challenges, particularly those involving long-range context.

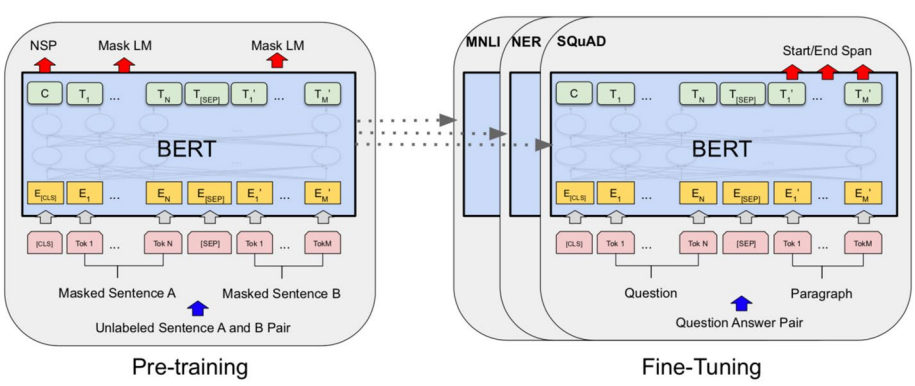

Fig. 4 BERT pre-training and fine-tuning procedures [29]

3.3 PaLM

The Pathways Language Model (PaLM), introduced by Chowdhery et al. [34], represents a state-of-the-art LLM with 540 billion parameters. Using Google's Pathways system, PaLM efficiently scales its model size while maintaining computational efficiency. Its architecture incorporates a Mixture of Experts (MoE) mechanism, where only a subset of experts processes each input, thereby reducing computational costs. The MoE layer is formalized as:

$$y = \sum_{i=1}^{k} G(x)_i E_i(x), \tag{3}$$

where x represents input, $G(x)_i$ is a gating function, E_i denotes the i-th expert, and y is the model output [35]. Furthermore, PaLM generalizes across diverse tasks with minimal examples by using contextual information from pretraining. This is represented as:

$$P(y|x, D) = \sum_{i=1}^{n} \alpha_i P(y|x, D_i), \tag{4}$$

where x is the input, D is the dataset, and α_i weights training examples [36]. These capabilities enable PaLM to perform well in reasoning, commonsense tasks, and mathematical problem-solving. The latest iteration, PaLM 2, scales up to even larger model sizes and incorporates advanced fine-tuning techniques to enhance its multilingual and reasoning capabilities [37].

3.4 LLaMA

The Large Language Model Meta AI (LLaMA), introduced by Touvron et al. [38], aims to deliver high performance with fewer computational resources. Ranging from 7 billion to 65 billion parameters, LLaMA focuses on data quality and model optimization over scale. Its architecture relies on the standard Transformer model but incorporates enhancements to improve efficiency, such as prioritizing high-quality training data [39].

LLaMA's self-attention mechanism allows the model to capture long-range dependencies while maintaining computational efficiency [40]. By effectively curating its training dataset, LLaMA achieves robust performance across diverse NLP benchmarks [39]. The latest versions, including LLaMA 2, LLaMA 3, LLaMA 3.1, LLaMA 3.2, and LLaMA 3.3, extend the parameter range to include models with up to 70 billion parameters and feature fine-tuned variants designed for specific applications.

3.5 T5

The Text-to-Text Transfer Transformer (T5), introduced by Raffel et al. [41], unifies all NLP tasks under a single text-to-text framework, transforming both inputs and outputs into text sequences. This abstraction allows a single model architecture to be used for a wide variety of tasks, including translation, summarization, and classification. T5 is based on the Transformer encoder-decoder structure, where the encoder processes the input and the decoder produces the output. It leverages the attention mechanism defined in Eq. 1 for capturing dependencies across tokens. T5 also adopts a span corruption objective during pretraining, masking spans of text and training the model to reconstruct them. The associated loss function is given by:

$$\mathcal{L}_{\text{mask}} = -\sum_{i=1}^{n} \log P(x_i | x_{\setminus i}, \theta), \tag{5}$$

where x_i is a masked token, $x_{\setminus i}$ are the unmasked tokens, and θ represents model parameters. The enhanced version, T5.1.1, introduces improvements such as pre-layer normalization and Adafactor optimization. Additionally, the multilingual variant mT5 extends T5's capabilities to over 100 languages, increasing its applicability in global NLP tasks [42].

3.6 Gemini

Gemini, developed by Google DeepMind [43], represents a multimodal LLM designed to integrate text and visual data seamlessly. Built upon the architectural foundation of PaLM, Gemini incorporates cross-modal attention mechanisms to enable reasoning across diverse modalities, such as text, images, and even structured data. The architecture of Gemini builds on the Transformer framework but incorporates a specialized cross-modal attention layer, where representations from different modalities are aligned [43]. This mechanism can be mathematically represented as:

$$\text{CrossAttention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}} + A_{\text{modality}}\right) V, \tag{6}$$

where A_{modality} is a modality-specific alignment matrix that adjusts the attention scores based on the modality of the input (e.g., text or image). Furthermore, the latest iterations, Gemini 1.5 and Gemini 2.0, introduce a hierarchical attention mechanism to dynamically adjust the weight of textual and visual components, enabling task-specific prioritization. This improvement enhances Gemini's ability to reason over complex multimodal tasks. Additionally, Gemini employs the mixture-of-experts approach, described in Eq. 3, to scale computational efficiency. In this framework, only a subset of experts is activated for each input, reducing resource demands.

3.7 DeepSeek-R1

DeepSeek-R1 was developed by the Chinese AI startup DeepSeek in early 2025 [44]. The model is part of a new wave of LLMs that aim to improve performance in complex reasoning tasks, particularly in mathematical and scientific domains. It builds upon the Mixture of Experts architecture, which enables dynamic expert routing to optimize computational efficiency. A key innovation in DeepSeek-R1 is its use of Reinforcement Learning with Reasoning Feedback (RLRF), a refinement over conventional Reinforcement Learning with Human Feedback (RLHF). While RLHF typically rewards outputs that align with subjective human preferences, RLRF introduces objective, structured signals based on logical soundness, correctness of intermediate steps, and coherence in multi-hop reasoning. These signals are generated via an automated verifier or teacher model, which evaluates the deductive consistency of model-generated outputs.

To optimize under RLRF, DeepSeek-R1 adopts Group Relative Policy Optimization (GRPO) [45–47], a novel reinforcement learning algorithm designed to enhance sample efficiency while reducing computational overhead. For each input query q, GRPO samples a group of outputs $\{o_1, o_2, \ldots, o_G\}$ from the previous policy $\pi_{\theta_{\text{old}}}$. These outputs

are evaluated using a reasoning-specific reward function, and the policy is updated by maximizing the GRPO objective.

$$J_{\mathrm{GRPO}}(\theta) = \mathbb{E}\left[\sum_{i=1}^{G} \min\left(\frac{\pi_\theta(o_i|q)}{\pi_{\theta_{\mathrm{old}}}(o_i|q)} A_i, \mathrm{clip}\left(\frac{\pi_\theta(o_i|q)}{\pi_{\theta_{\mathrm{old}}}(o_i|q)}, 1-\epsilon, 1+\epsilon\right) A_i\right)\right] - \beta D_{\mathrm{KL}}(\pi_\theta || \pi_{\mathrm{ref}}), \quad (7)$$

where ϵ is a clipping hyperparameter (as in PPO), and β regulates the KL penalty to prevent divergence from a reference policy π_{ref}. The advantage estimate A_i used in the GRPO objective is calculated by normalizing the reward scores across the sampled outputs. Specifically, each advantage is computed as the standardized difference from the group mean: $A_i = \frac{r_i - \mu}{\sigma}$, where $\mu = \mathrm{mean}(\{r_1, \ldots, r_G\})$ and $\sigma = \mathrm{std}(\{r_1, \ldots, r_G\})$. This normalization helps ensure numerical stability and effective credit assignment during policy updates.

4 Training and adaptation techniques

The training and adaptation of LLMs involve a variety of methods designed to help these models learn complex linguistic patterns and effectively apply this knowledge to diverse tasks. This section covers important techniques ranging from foundational training methods to advanced mechanisms, such as in-context learning and retrieval-augmented generation.

4.1 Pre-training

Pre-training forms the foundation of LLM development as it enables models to learn linguistic patterns and semantic structures from vast, unlabeled datasets using self-supervised learning techniques [48]. Two dominant pre-training objectives are autoregressive modeling and masked language modeling [49]. Autoregressive modeling, employed by models such as GPT, predicts the next token in a sequence given the preceding tokens, making it suitable for text generation tasks. The loss function for autoregressive pre-training is expressed as:

$$\mathcal{L}_{pretrain} = -\sum_{i=1}^{n} \log P(x_i | x_{<i}, \theta), \quad (8)$$

where x_i is the token to be predicted, $x_{<i}$ represents preceding tokens, and θ denotes model parameters. In contrast, MLM, as employed by models such as BERT, involves masking random tokens in a sequence and training the model to predict them based on the surrounding context. This bidirectional approach captures dependencies in both forward and backward directions, thereby enhancing performance in tasks such as question answering and text classification [50]. Mathematically, the MLM loss function is:

$$\mathcal{L}_{MLM} = -\sum_{i=1}^{n} \log P(x_i | x_{\setminus i}, \theta), \quad (9)$$

where x_i represents the masked token and $x_{\setminus i}$ denotes the unmasked tokens [29]. Through these objectives, pre-training equips LLMs with a general understanding of language, thereby enabling their application across diverse NLP tasks.

4.2 Fine-tuning

Fine-tuning adapts pre-trained LLMs to specific downstream tasks by leveraging smaller, task-specific labeled datasets [51, 52]. Unlike pre-training, which generalizes language patterns, fine-tuning optimizes model performance for tasks, such as sentiment analysis or text classification. The fine-tuning process minimizes a supervised loss function, such as cross-entropy, defined as:

$$\mathcal{L}_{fine-tune} = -\sum_{i=1}^{n} y_i \log P(y_i|x_i, \theta), \tag{10}$$

where y_i is the true label, and $P(y_i|x_i, \theta)$ represents the predicted probability of the label for input x_i. Furthermore, fine-tuning typically requires fewer resources than pre-training, as it focuses on task-specific adaptations rather than building general language understanding [52]. This method also enables domain adaptation, which makes LLMs pre-trained on general corpora robust in specialized domains.

4.3 Transfer learning

Transfer learning enables LLMs to leverage knowledge acquired during pre-training for new tasks or domains with minimal additional training [53, 54]. By reusing pre-trained representations, models can generalize to tasks requiring limited labeled data. For example, a model trained on general corpora can be fine-tuned for domain-specific tasks, such as sentiment analysis or legal document processing, thereby reducing computational costs.

Unlike fine-tuning, which typically involves updating most or all of the model's parameters to optimize it for a specific task, transfer learning often focuses on reusing the pre-trained model with minimal parameter updates, or even keeping them frozen while adding task-specific layers. This distinction allows transfer learning to be computationally efficient, particularly in scenarios where labeled data is scarce.

Furthermore, this approach is effective in cross-lingual scenarios, where an English-trained LLM can adapt to other languages with minimal fine-tuning. Transfer learning enhances the versatility of LLMs, making them applicable across diverse tasks and reducing the need for extensive labeled datasets.

4.4 Reinforcement learning with human Feedback

Reinforcement Learning with Human Feedback (RLHF) aligns LLM outputs with human preferences by incorporating human evaluations into the training loop [55, 56]. In RLHF, human annotators rank multiple model outputs, and these rankings define a reward function that the model seeks to maximize. The objective function for RLHF is:

$$\mathcal{L}_{RLHF} = \mathbb{E}_{\pi_\theta}[r(s, a)], \tag{11}$$

where $r(s, a)$ represents the reward for action a in state s, and π_θ denotes the model's policy [57]. RLHF has been instrumental in improving conversational AI systems by aligning responses with user expectations, reducing inappropriate outputs, and addressing ethical concerns. This technique also enhances fairness by incorporating human oversight, making it crucial for high-stakes applications, including healthcare and law.

4.5 Few-shot, zero-shot, and in-context learning

Few-shot learning refers to the ability of LLMs to generalize to new tasks after seeing only a few task-specific examples, while zero-shot learning involves performing tasks without any explicit examples provided during inference. In-context learning is the mechanism underlying these approaches, wherein the model uses patterns presented in the input prompt context to infer appropriate outputs [58].

Models like GPT-3 and GPT-4 have demonstrated remarkable capabilities in these scenarios. For instance, GPT-3 achieved approximately 71.8% accuracy on the SuperGLUE benchmark under few-shot conditions, significantly narrowing the performance gap compared to fully fine-tuned models [8]. GPT-4 further enhanced this performance, demonstrating higher accuracy and better generalization across diverse linguistic and reasoning tasks [15]. Google's PaLM model similarly excelled, outperforming GPT-3 on several language understanding and reasoning tasks, such as achieving state-of-the-art results on the BIG-bench benchmark [34].

4.6 Retrieval-augmented generation

Retrieval-Augmented Generation (RAG) combines LLM capabilities with retrieval mechanisms to generate contextually accurate and factually grounded responses [59]. Instead of relying solely on pre-trained knowledge, RAG retrieves relevant documents or snippets from external knowledge bases, such as Wikipedia or specialized datasets, during inference. The retrieved information is then integrated into the response generation process. The probability of generating text in RAG is expressed as:

$$P(x|z) = P(x|z,q)P(z|q), \tag{12}$$

where x is the generated text, z is the retrieved document, and q is the input query [60]. RAG is particularly effective for knowledge-intensive tasks such as answering questions and summarizing technical documents. Additionally, RAG addresses challenges such as hallucinations associated with LLMs and improves the factual accuracy of outputs [61].

4.7 Prompt engineering techniques and their influence on LLM performance

Prompt engineering refers to designing and refining prompts to effectively elicit desired responses from LLMs without extensive retraining. It has become a critical technique, significantly influencing the performance of LLMs in various tasks including text generation, reasoning, classification, and ethical alignment [62]. Techniques such as zero-shot, few-shot, and chain-of-thought prompting have been particularly influential. Zero-shot prompting involves providing LLMs with a clear task description without any explicit examples, allowing models to generalize effectively. Formally, given an input x and a prompt p, zero-shot generation aims to maximize the conditional probability of the output y:

$$P(y|x,p) = \prod_{i=1}^{n} P(y_i|y_{<i},x,p) \tag{13}$$

where y_i represents each generated token, and $y_{<i}$ denotes tokens generated thus far [63]. In contrast, few-shot prompting provides a limited number of demonstrations within the prompt itself, improving the model's task-specific adaptation capability:

$$P(y|x,p,D) = \prod_{i=1}^{n} P(y_i|y_{<i},x,p,D) \tag{14}$$

where D represents a set of few-shot demonstrations [8]. Meanwhile, chain-of-thought (CoT) prompting enhances model performance, especially in complex reasoning tasks, by decomposing prompts into a sequence of intermediate reasoning steps [64]. Mathematically, the CoT generation process is described as:

$$P(y,r|x,p) = P(r|x,p) \cdot P(y|r,x,p) \tag{15}$$

where r is the set of intermediate reasoning steps produced by the model before the final answer y. Recent research highlights the pivotal role prompt engineering plays in enhancing the robustness, interpretability, and ethical alignment of LLM outputs. Techniques such as self-consistency prompting have been shown to reduce model hallucinations and improve factual correctness by generating multiple reasoning paths and selecting consistent answers [65]. Additionally, research on prompt engineering for specific applications, such as detecting sexism [66] or optimizing energy usage scheduling [6], illustrates the breadth and practical importance of well-engineered prompts.

4.8 Text preprocessing methods

Text preprocessing plays a crucial role in enhancing the performance, efficiency, and accuracy of LLMs within the training pipeline. Key preprocessing techniques include tokenization, lemmatization, stemming, and stop-word removal. Among these, tokenization serves as an initial essential step, where text is segmented into tokens or meaningful units. This step directly affects vocabulary construction and embedding quality, thereby significantly influencing a model's capability to accurately interpret context and meaning [67].

Subsequent steps such as lemmatization and stemming focus on normalizing word variants, thus reducing vocabulary size and computational complexity. Lemmatization converts words to their base or dictionary forms while preserving their semantic integrity, making it particularly useful for tasks where precise meaning is essential. Conversely, stemming simplifies words by truncating them to their root forms based on heuristic rules, often resulting in some loss of semantic clarity [68]. The choice between lemmatization and stemming typically depends on task-specific requirements, balancing between semantic accuracy and computational efficiency.

Additionally, stop-word removal further refines the dataset by eliminating commonly occurring yet semantically insignificant words such as *"the," "is,"* and *"at"*. This step effectively reduces noise within the data, thereby enhancing the meaningful representation of the input text [69]. Handling special characters and symbols is also vital to mitigate inconsistencies and noise across datasets. Recent research demonstrates the significance of well-designed preprocessing pipelines, demonstrating their positive impact on the performance and generalization capabilities of transformer-based models.

5 Evaluation of LLMs and benchmark datasets

This section explores the primary evaluation metrics and widely used benchmark datasets in assessing the effectiveness of LLMs.

5.1 Evaluation metrics

Evaluating the performance of LLMs is crucial to understanding their capabilities and limitations across various tasks. Unlike traditional models, LLMs must be evaluated across a diverse set of benchmarks, taking into account both their linguistic proficiency and their ability to generalize across domains [70]. Below, we discuss the most commonly used evaluation metrics.

5.1.1 Perplexity

Perplexity is a measure of how well a language model predicts a sequence of words [71]. It is widely used to assess the fluency and predictability of language models. Mathematically, perplexity is the exponentiation of the average negative log-likelihood per word:

$$Perplexity\ (P) = \exp\left(-\frac{1}{N}\sum_{i=1}^{N}\log P(x_i|x_{<i})\right) \tag{16}$$

where N is the number of words in the sequence, and $P(x_i|x_{<i})$ is the model's predicted probability for the next word given the preceding words [5, 71]. A lower perplexity score indicates that the model is better at predicting the next word, making it more fluent in text generation. However, perplexity alone cannot evaluate the model's performance in specific tasks, including summarization or reasoning.

5.1.2 Exact match and F1-Score

These metrics are useful in tasks such as question answering and information retrieval, where the goal is to match specific reference answers [72]. The exact match (EM) score calculates the percentage of predictions that match the reference answers exactly. It is defined as:

$$EM = \frac{1}{N}\sum_{i=1}^{N}\mathbb{I}[\hat{y}_i = y_i] \tag{17}$$

where N is the total number of examples, $\hat{y}_i$ is the model's predicted answer, y_i is the ground truth answer, and $\mathbb{I}[\cdot]$ is the indicator function, which equals 1 if the predicted answer matches the true answer and 0 otherwise [73]. Meanwhile, the F1 score is the harmonic mean of precision and recall [74].

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \tag{18}$$

where, Precision is a measure of how many of the model's predicted answers are actually correct, focusing on the accuracy of the predictions made. Recall, on the other hand, measures how many of the correct answers from the reference set are successfully identified by the model, highlighting its ability to find all relevant answers. The F1-score provides a broader view of performance by balancing how many correct answers are generated and how relevant those answers are to the query.

5.1.3 Bilingual evaluation understudy

Bilingual Evaluation Understudy (BLEU) is widely used in machine translation and text summarization to evaluate how closely a generated text matches a reference text [75]. It

works by comparing the overlap of n-grams between the generated text and the reference. The BLEU score is calculated as:

$$BLEU = BP \cdot \exp\left(\sum_{n=1}^{N} w_n \log p_n\right) \tag{19}$$

where p_n is the precision of n-grams, w_n is a weight for each n-gram, and BP is the brevity penalty, introduced to penalize short translations. BLEU provides a good measure of linguistic accuracy but may not always capture semantic correctness, which is essential in tasks such as creative writing or abstract summarization.

5.1.4 Recall-oriented understudy for gisting evaluation

Recall-Oriented Understudy for Gisting Evaluation (ROUGE) is another widely used metric, mainly in text summarization tasks. It measures the overlap of n-grams between the generated and reference texts, focusing on recall rather than precision [76]. ROUGE has different variants, but two of the most common are ROUGE-N and ROUGE-L [77]. ROUGE-N measures the overlap of n-grams, and ROUGE-L measures the longest common subsequence (LCS) between the generated and reference text. The LCS ensures that the generated summary preserves the structure and flow of the reference summary [78]. ROUGE-N is calculated as:

$$\text{ROUGE-N} = \frac{\sum_{\text{n-gram}\in\text{ref}} \min\left(f(\text{n-gram in ref}), f(\text{n-gram in hyp})\right)}{\sum_{\text{n-gram}\in\text{ref}} f(\text{n-gram})} \tag{20}$$

$f(\text{n-gram in ref})$ and $f(\text{n-gram in hyp})$ represent the frequency of a specific n-gram in the reference text and the hypothesis text, respectively. The numerator calculates the sum of the minimum frequencies for all n-grams found in both the reference and hypothesis, ensuring that the overlap of n-grams is considered. The denominator computes the total frequency of all n-grams in the reference text. This formulation allows ROUGE-N to measure how well the n-grams in the generated text (hypothesis) match those in the reference text. Meanwhile, ROUGE-L is computed as follows:

$$\text{ROUGE-L} = \frac{LCS(\text{ref}, \text{hyp})}{\ell(\text{ref})} \tag{21}$$

where $LCS(\text{ref}, \text{hyp})$ represents the length of the longest common subsequence between the reference summary (ref) and the generated summary (hyp), and $\ell(\text{ref})$ denotes the length of the reference summary [79].

5.1.5 Accuracy

Accuracy is a basic yet essential metric for classification tasks, such as sentiment analysis or named entity recognition (NER). It measures the proportion of correctly predicted labels to the total number of labels [80]. The formula for accuracy can be expressed as:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{22}$$

where TP represents True Positives and TN is True Negatives, while FP and FN denotes False Positives and False Negatives, respectively.

5.1.6 Calibration, factuality, and truthfulness

Beyond traditional metrics, recent emphasis has shifted toward calibration, factuality, and truthfulness to address reliability and hallucination concerns. Calibration evaluates how well model-generated probabilities reflect actual correctness, enhancing interpretability and trust [81]. Factuality and truthfulness metrics specifically assess a model's capability to produce accurate, verifiable content. For example, TruthfulQA benchmark explicitly measures truthfulness, challenging models with questions prone to common misconceptions or inaccuracies [82]. Integrating these metrics provides a comprehensive evaluation framework that addresses the critical limitations of LLMs in generating trustworthy and factual outputs.

5.2 Benchmark datasets

Benchmark datasets play an essential role in assessing the performance and capabilities of LLMs across various NLP tasks. Commonly used benchmarks such as GLUE [83], SQuAD [84], SuperGLUE [85], MS MARCO [86], and MMLU [87] enable comprehensive evaluation of language understanding, reasoning, and question-answering skills. Each of these datasets targets specific linguistic and cognitive capabilities, providing a standardized basis for comparing LLMs.

However, the effectiveness of these benchmarks significantly depends on dataset quality, curation, and preprocessing methods. Data filtering and noise removal processes are particularly critical, as they directly affect the reliability and validity of model evaluations. For instance, inadequate filtering practices can introduce contamination and data leakage, leading to artificially inflated performance metrics and compromised applicability in real-world scenarios [88]. Recent literature shows the importance of rigorous data cleaning techniques and advocates for greater transparency in dataset creation processes to address these concerns [89].

6 Responsible AI and governance frameworks

The rapid advancement of LLMs has brought significant benefits across various sectors, but it has also raised important ethical and governance concerns. As LLMs are deployed in high-stakes areas such as healthcare, law, and public administration, ensuring their responsible development and use is critical. This section explores the frameworks that guide responsible AI, including transparency, accountability, fairness, and the evolving landscape of AI governance.

6.1 Ethical considerations for responsible AI

One of the primary concerns surrounding LLMs is the ethical impact of their deployment in areas where their decisions or recommendations can have real-world consequences. Several important ethical principles form the basis of responsible AI development, including:

6.1.1 Fairness and non-discrimination

LLMs are trained on large datasets scraped from the internet, which often contain biases related to race, gender, and socioeconomic status. If these biases are not properly mitigated, LLMs can perpetuate or even exacerbate existing inequalities [90]. Fairness in AI requires models to be developed and tested rigorously to avoid biased outputs that could

harm marginalized groups. Frameworks such as the Fairness in ML approach advocate for the development of methods to audit and ensure fairness in model outputs.

6.1.2 Transparency and explainability

The "black box" nature of LLMs, where it is difficult to understand how specific outputs are generated, raises concerns about transparency and accountability. Explainability is crucial in high-stakes fields such as legal decision-making and healthcare, where understanding the rationale behind a decision is essential for building trust [91]. Researchers and organizations are working on interpretability tools, such as attention visualization and saliency maps, to make LLMs more transparent.

6.1.3 Accountability

Developers and users of LLMs must take responsibility for the potential harm their models could cause [92]. This includes ensuring that models are used for intended purposes and implementing measures to prevent misuse. Legal frameworks are beginning to emerge around accountability, such as the EU AI Act, which imposes requirements for risk management and oversight on AI developers.

6.1.4 Privacy and data protection

As LLMs are trained on massive datasets that can contain sensitive or personal information, privacy concerns are paramount. Responsible AI frameworks must ensure that data protection regulations, such as the General Data Protection Regulation (GDPR), are strictly followed. Techniques such as differential privacy are being used to minimize the risk of sensitive data exposure while still allowing LLMs to perform effectively [93].

6.2 Governance frameworks for AI

To address the ethical challenges posed by LLMs, governance frameworks have been developed to guide the responsible use of AI [65]. These frameworks are designed to create guidelines for the safe and equitable deployment of AI systems, ensuring compliance with ethical standards and regulatory requirements.

6.2.1 International governance frameworks

Globally, several organizations and coalitions have developed frameworks aimed at enabling responsible AI development:

- OECD AI Principles: The Organization for Economic Cooperation and Development (OECD) introduced a set of AI principles in 2019, which have since been adopted by more than 40 countries [94]. These principles emphasize inclusive growth, human-centered values, and transparency in AI development. Importantly, they call for both national and international cooperation in establishing AI regulations.
- European Union AI Act: The EU AI Act is a landmark regulatory framework that categorizes AI applications into different risk levels, from minimal to high risk. High-risk applications, such as those in healthcare, must comply with stringent requirements, including transparency, robustness, and human oversight [95, 96]. The EU AI Act is one of the most comprehensive frameworks, providing detailed guidelines for companies developing or deploying AI in the European market.

- UNESCO Recommendations on AI Ethics: In 2021, UNESCO introduced a framework for the ethical use of AI, focusing on ensuring that AI development is aligned with human rights, democratic values, and sustainability [97]. UNESCO's framework emphasizes that AI should be used to benefit all humanity, not just privileged groups, and it highlights the importance of protecting cultural diversity and human dignity.

6.2.2 Corporate and institutional governance

Many leading technology companies and research institutions have also implemented their own governance structures to ensure the responsible use of AI:

- Google's AI Principles: Google has developed a set of AI principles that guide their development and deployment of LLMs. These principles emphasize avoiding bias, ensuring privacy, and being accountable for AI's impact. Google has also committed to using AI responsibly by auditing its models for fairness and explainability, and by releasing transparency reports that outline the ethical implications of its AI applications [98].
- OpenAI's Safety Guidelines: OpenAI has put forward its own governance mechanisms to ensure that LLMs such as GPT-4 are deployed safely. This includes a focus on human oversight, transparency, and preventing misuse in harmful contexts. OpenAI emphasizes the importance of continuous model evaluation and the use of RLHF to align models with human values [99].
- Partnership on AI: The Partnership on AI is a multi-stakeholder organization involving tech companies, academia, and civil society groups, aiming to establish industry standards for responsible AI development [100]. The partnership focuses on building transparency and accountability into AI systems while promoting collaboration across sectors to solve ethical challenges.

7 Emerging applications and innovations

LLMs have traditionally been associated with tasks such as text generation, translation, and summarization. However, recent advancements have expanded their utility to more specialized and innovative applications across various domains. In this section, we explore these emerging use cases.

7.1 Code generation and debugging

One of the most groundbreaking applications of LLMs is their ability to assist with code generation and debugging. Models such as OpenAI's Codex, a derivative of GPT-3, have been trained specifically to understand and generate programming languages [101]. Codex powers GitHub Copilot, a tool that helps developers by auto-completing code, suggesting functions, and even writing entire code snippets based on natural language descriptions. This integration of LLMs into the software development process enables faster prototyping, reduces the manual effort required for repetitive coding tasks, and assists in debugging by identifying potential errors in real time. Recent studies have evaluated Codex's effectiveness in both generating and correcting code. For instance, Zheng et al. [102] demonstrate that Codex performs well on common programming tasks, particularly in languages such as Python, achieving high accuracy in functional code generation.

However, there are limitations. For example, Pearce et al. [103] have identified cases where Codex can generate insecure code, suggesting that while LLMs reduce workload, developers still need to exercise caution when using these models in sensitive contexts. Further research by Vaithilingam et al. [104] explores user expectations of AI-assisted programming tools such as Copilot, noting that although these models enhance efficiency, developers often over-rely on them, resulting in the propagation of subtle bugs that the models fail to detect.

Recent works argue that LLM-based tools are revolutionizing software engineering by making programming more accessible to non-experts. For example, Nguyen et al. [105] highlight how Copilot allows individuals with limited coding experience to participate in software development, effectively democratizing coding by lowering the barrier to entry. They suggest that LLMs enable rapid prototyping and allow developers to focus on higher-level architectural decisions rather than mundane coding tasks. Additionally, Jiang et al. [106] have found that novice developers can significantly benefit from LLM assistance, as these models can offer real-time suggestions and generate error-free code snippets, improving both productivity and learning outcomes.

Conversely, some researchers argue that while LLMs offer significant productivity gains, they may introduce new challenges related to software quality and developer reliance. Zhang et al. [107] suggest that the use of LLMs for code generation can lead to over-reliance on AI-generated solutions, potentially reducing the depth of understanding that developers have about the underlying logic of their code. Moreover, they emphasize that developers might prioritize speed over accuracy, potentially introducing subtle bugs that could go unnoticed. These differing perspectives highlight the need for further studies on the long-term impact of LLM-based code generation tools on software development practices and code quality.

7.2 Legal and contractual analysis

In the legal field, LLMs are transforming the way law firms and legal professionals handle time-consuming tasks such as legal document drafting, contract analysis, and due diligence as shown by Katz et al.[108]. Traditionally, these tasks required a significant amount of manual labor, often leading to errors or inefficiencies. However, LLMs are now capable of processing large volumes of legal texts, extracting relevant clauses, and even suggesting modifications to legal documents based on past case law. This not only improves operational efficiency but also reduces the risk of human error in legal drafting.

Several recent studies have evaluated the use of LLMs in the legal domain. For example, Aejas et al. [109] found that LLMs, when trained on domain-specific data, are highly effective in tasks such as contract clause extraction and classification. Their work demonstrated that models, such as BERT, when fine-tuned on legal corpora, outperform traditional NLP techniques in accurately identifying legal terms and conditions, which is critical for contract review and compliance monitoring. Moreover, these tools are being deployed to handle large-scale contract reviews in mergers and acquisitions, reducing the time required to identify potential legal risks.

LLMs have also been shown to assist in case law analysis, a task that typically requires legal professionals to search through vast databases of legal precedents. Padiu et al. [110] demonstrated how models such as GPT-3 are being adapted to generate legal summaries, extract insights from previous rulings, and predict potential outcomes of ongoing

cases. This predictive capability enables legal professionals to make better-informed decisions, not only speeding up the research process but also improving the accuracy of legal predictions. Furthermore, it allows law firms to provide more data-driven strategies for litigation and negotiation.

However, concerns remain regarding the interpretability and transparency of LLMs in legal contexts. Studies such as Nay et al. [111] highlight the need for greater transparency in AI-generated legal outputs. While LLMs can significantly enhance the efficiency of legal services, there is a need for continuous validation to ensure that the AI-generated text aligns with legal standards and does not inadvertently introduce biased or incorrect interpretations of the law. The issue of bias in LLMs is of particular concern, as it can affect the fairness of outcomes and erode trust in AI-assisted legal tools.

Lastly, Yang et al. [112] and Harasta et al. [113] indicated that the automation of legal tasks through LLMs is expected to reduce the burden on legal professionals and free up time for higher-level strategic work, such as litigation planning and negotiation.

7.3 Healthcare diagnostics and research

In healthcare, LLMs are increasingly playing a crucial role in diagnostics, medical research, and clinical decision-making [114, 115]. LLMs such as MedPaLM have been developed to process and interpret medical records, summarize diagnostic findings, and assist healthcare professionals in identifying potential treatment options as shown by Singhal et al. [116]. These models are fine-tuned on large medical datasets to ensure they can handle complex healthcare data, such as electronic health records (EHRs), radiology reports, and genetic information. This has enabled clinicians to make faster, more informed decisions, especially in time-sensitive scenarios. Similarly, [117] used Med-PaLM 2 for liver cancer question answering.

Furthermore, one of the main advantages of large language models (LLMs) in healthcare is their ability to assist with diagnostics. Models such as BioGPT have been shown to improve the accuracy of medical diagnoses by providing context-aware recommendations and flagging potential errors in physician input, as demonstrated by Luo et al. [118]. Additionally, in radiology, LLMs can assist in interpreting medical images and providing detailed reports, reducing the workload on radiologists while maintaining high accuracy levels. In a recent study, Park et al. [119] explored how LLMs can be used to assist with interpreting mammograms. Their model was able to highlight subtle details that were initially missed by human radiologists. This is especially important for improving early detection rates of diseases such as cancer, where timely diagnosis can save lives.

LLMs are also revolutionizing medical research by helping scientists navigate the vast amounts of biomedical literature available today. Researchers often struggle to keep up with the rapidly expanding volume of publications, but LLMs are now being used to automatically scan and synthesize findings from multiple studies. For instance, Chakrabort et al. [120] employed a ChatGPT-based model for biomedical text processing to assist researchers in identifying novel drug interactions, and Zamora-Resendiz et al. [121] employed LLMs for disease progression, aiming to uncover hidden patterns in patient outcomes. By automating these tasks, LLMs enable researchers to focus on more complex aspects of medical innovation, speeding up the discovery process and contributing to advancements in personalized medicine.

7.4 Creative and artistic tools

LLMs also find applications in creative domains, where they assist in generating artistic content such as poetry, music lyrics, and even visual art descriptions. By understanding the context and style of input prompts, these models can mimic various artistic forms, aiding in brainstorming and content creation. For instance, Chun and Elkins [122] employed GPT-4 to analyze and generate stories. Chen et al. [123] explored an LLM called HoLLMwood for screenwriting, showing promising results. This blend of AI and creativity opens new avenues for collaboration between human artists and machines.

In the gaming industry, LLMs are being used to create dynamic dialogue for non-playable characters (NPCs), enhancing the immersive experience for players. Instead of pre-scripted, repetitive dialogue, LLMs allow NPCs to respond in real time, adjusting their responses based on the player's actions and in-game context. This development can lead to more engaging and personalized gaming experiences as shown by Pirrone and D'Ulizia [124]. Recent research works by Zhu et al. [125] and Zhang et al. [126] have shown that LLMs can also aid in the design process of game worlds, offering suggestions for plot developments, character backstories, and even gameplay mechanics, which significantly reduces the time developers spend on these tasks.

Meanwhile, LLMs are also gaining prominence in visual arts. Liu et al. [127] developed an LLM-based approach that can assist artists by generating descriptive art concepts and even detailed visual scene descriptions. Ko et al. [128] proposed a method for text-to-image for visual artists' creative works. This collaborative approach between text-generation models and image-generation AI opens up new possibilities for artists looking to experiment with styles, settings, and concepts they may not have considered otherwise. Furthermore, in the music industry, Gardner et al. [129] developed an LLM called Llark for music generation while Liu and Liu [130] developed an LLM for generating musician lyrics in Mandarin. These innovations show how LLMs can revolutionize the creative industry.

7.5 Real-time cybersecurity

One of the more innovative uses of LLMs is in the field of cybersecurity, where these models are employed to detect and prevent cyber threats in real-time. LLMs can process large amounts of data related to network traffic, identifying anomalies that could signal potential security breaches. These models are robust at identifying patterns in unstructured data, which allows them to detect unusual activity indicative of malicious behavior, such as unauthorized access or data exfiltration attempts as shown by Sarker [131].

LLMs are also becoming effective tools in combating phishing attacks by analyzing email content, identifying suspicious links, and flagging messages that exhibit characteristics of known phishing strategies. A study by Bethany et al. [132] demonstrated that LLMs could help detect and respond to phishing attacks faster than traditional rule-based systems. Additionally, Elouardi et al. [133] developed an approach that integrates LLMs with ML algorithms to automatically classify malware, which can enhance an organization's ability to thwart cyber threats.

LLMs have also been used to automate incident response. They achieve this by analyzing threat intelligence and correlating data from multiple sources to generate automated responses, which can mitigate attacks before they escalate. For instance, Shafee et al. [134] and Zhang et al. [135] showed that LLMs can reduce the time needed for threat

detection and response by automating common cybersecurity tasks, such as log analysis, generating alerts, and isolating compromised systems. This capability is especially valuable in modern cybersecurity environments, where the speed and sophistication of attacks are increasing rapidly.

Furthermore, LLMs are enhancing the scalability of cybersecurity measures, enabling organizations to monitor larger and more complex networks with fewer human resources. Research by Hassanin and Moustafa et al. [136] has shown that LLMs are capable of processing real-time data streams and generating insights that help security teams respond to evolving threats more efficiently. This proactive approach to cybersecurity is valuable in today's environment, where the volume and sophistication of cyber threats continue to grow.

7.6 Financial analysis

One of the most innovative applications of LLMs in finance is their use in real-time financial analysis. These models can process vast amounts of market data, identifying trends and anomalies that inform investment strategies and risk management decisions. LLMs are adept at analyzing unstructured data from news articles, social media, and financial reports, enabling them to recognize patterns that might indicate market shifts or emerging opportunities, as demonstrated by Liu et al. [137].

LLMs are also becoming essential tools in sentiment analysis, where they evaluate the tone and context of financial news and social media posts. Wu et al. [138] showcased how BloombergGPT, a 50 billion parameter language model, supports various financial tasks, including sentiment assessment. This capability allows financial analysts to gauge market sentiment quickly, enhancing decision-making processes in fast-paced environments.

Moreover, LLMs facilitate the automation of financial reporting and analysis. Han et al. [139] introduced XBRL-Agent, which improves the interpretation and accessibility of financial reports by leveraging external tools. This approach significantly enhances accuracy in domain and numeric queries, streamlining the financial analysis process and reducing the time required to generate insights.

In market risk assessment, LLMs can analyze historical data and compute risk metrics to forecast potential market downturns. Lefort et al. [140] demonstrated that ChatGPT could compute sentiment scores from Bloomberg Market Wraps, establishing a statistically significant correlation between sentiment and future equity market returns. This capability aids analysts in making informed decisions regarding risk exposure.

LLMs are continually enhancing the scalability and efficiency of financial analysis, allowing institutions to monitor and analyze larger datasets with fewer resources. Bhatia et al. [141] introduced FinTral, a multimodal financial LLM that integrates text, numbers, tables, and images, demonstrating superior performance in real-time analysis. This proactive approach to financial analysis is invaluable in today's rapidly evolving market landscape, where timely and accurate insights are crucial for maintaining a competitive edge.

Table 1 shows a summary of the emerging LLM applications in different domains.

Table 1 Summary of emerging applications and innovations of llms across various domains.

Application domain	References	Year	Methods and applications
Code generation and debugging	Chen et al. [101]	2021	Codex powers GitHub Copilot, enabling code autocompletion, function suggestions, and debugging, enhancing productivity.
	Zheng et al. [102]	2023	Demonstrated Codex's high accuracy in functional code generation for common programming tasks.
	Pearce et al. [103]	2022	Identified risks of insecure code generated by LLMs, emphasizing cautious use in sensitive contexts.
	Vaithilingam et al. [104]	2022	Explored user expectations, highlighting over-reliance on LLMs and the propagation of subtle bugs.
	Nguyen et al. [105]	2024	Discussed democratization of coding through LLMs, enabling non-experts to participate in software development.
	Jiang et al. [106]	2024	Highlighted benefits for novice developers, improving productivity and learning outcomes through real-time suggestions.
	Zhang et al. [107]	2023	Examined potential over-reliance on LLM-generated code and its implications on software quality.
Legal and contractual analysis	Katz et al. [108]	2024	Streamlined legal document review, contract analysis, and clause extraction, improving efficiency and reducing errors.
	Aejas et al. [109]	2024	Demonstrated the effectiveness of domain-specific LLMs in identifying legal terms and conditions.
	Padiu et al. [110]	2024	Showed LLMs' ability to generate legal summaries, analyze precedents, and predict case outcomes.
	Nay et al. [111]	2024	Highlighted transparency concerns in AI-generated legal outputs, emphasizing validation.
	Yang et al. [112]	2024	Predicted automation of legal tasks to enable higher-level strategic work, such as litigation planning.
	Harasta et al. [113]	2024	Discussed the potential for LLMs to optimize litigation planning and negotiation workflows.
Healthcare diagnostics and research	Kim et al. [114]	2024	Improved medical diagnostics through context-aware recommendations and reduced radiologist workload.
	Singhal et al. [116]	2023	Leveraged MedPaLM for interpreting complex medical records and summarizing diagnostic findings.
	Qian et al. [117]	2024	Employed MedPaLM 2 for liver cancer question answering.
	Luo et al. [118]	2022	Enhanced diagnostic accuracy using BioGPT for context-aware recommendations.
	Park et al. [119]	2024	Demonstrated LLMs' ability to assist radiologists by identifying subtle details in mammograms.
	Chakraborty et al. [120]	2023	Applied LLMs to biomedical literature analysis for drug interaction identification.
	Zamora-Resendiz et al. [121]	2024	Used LLMs to study disease progression and uncover hidden patient data patterns.
Creative and artistic tools	Chun and Elkins [122]	2023	Generated creative stories using GPT-4 for analysis and storytelling applications.
	Chen et al. [123]	2024	Developed HoLLMwood, an LLM for screenwriting and collaborative story generation.
	Pirrone and D'Ulizia [124]	2024	Enabled dynamic NPC dialogue in gaming, enhancing immersive and personalized experiences.
	Zhu et al. [125]	2023	Assisted in game world design, offering plot, character, and gameplay suggestions.
	Liu et al. [127]	2024	Supported artists by generating descriptive art concepts and detailed visual scene descriptions.
	Gardner et al. [129]	2023	Created Llark, an LLM for music generation and innovative musical compositions.

Table 1 (continued)

Application domain	References	Year	Methods and applications
Real-time cybersecurity	Sarker [131]	2024	Used LLMs for real-time anomaly detection in network traffic to identify potential cyber threats.
	Bethany et al. [132]	2024	Aided phishing detection through email content analysis and suspicious link identification.
	Elouardi et al. [133]	2024	Integrated LLMs with ML algorithms to classify malware, improving cybersecurity.
	Zhang et al. [135]	2024	Automated incident response, reducing response times and improving organizational security efficiency.
	Shafee et al. [134]	2024	Demonstrated LLMs' role in correlating threat intelligence and automating threat mitigation.
Financial analysis	Liu et al. [137]	2023	Introduced FinGPT, an open-source framework that automates the collection and curation of real-time financial data from diverse internet sources, enabling the development of specialized financial large language models.
	Wu et al. [138]	2023	Created the BloombergGPT, a 50 billion parameter language model designed to support a wide range of financial-related tasks.
	Han et al. [139]	2024	Developed XBRL-Agent, an LLM-based agent framework designed to improve the interpretation and accessibility of XBRL financial reports by leveraging external tools such as retrievers and calculators, significantly enhancing accuracy in domain and numeric query tasks.
	Lefort et al. [140]	2024	Demonstrated the use of ChatGPT to compute sentiment scores from Bloomberg Market Wraps, establishing a statistically significant correlation between sentiment and future equity market returns, aiding in market risk assessment.
	Zhao et al.[142]	2024	Provided a comprehensive overview of LLM integration into financial tasks, including automating report generation and forecasting market trends, highlighting GPT-4's effectiveness across various financial tasks.
	Bhatia et al. [141]	2024	Introduced FinTral, a multimodal financial LLM based on Mistral-7B, integrating text, numbers, tables, and images. Demonstrated superior zero-shot performance, outperforming ChatGPT-3.5 in all tasks and GPT-4 in five out of nine tasks. Designed for real-time financial analysis and decision-making.

8 Proposed conceptual framework for categorizing LLMs

As LLMs continue to scale in size and capability, it becomes increasingly important to create a systematic framework for categorizing and understanding them. This section proposes a conceptual framework classifying LLMs based on three main dimensions: scalability, application domains, and ethical impact. The objective is to provide a structured taxonomy guiding future research, development, and governance of these models.

8.1 Scalability tiers

Scalability, primarily measured by the number of model parameters, significantly influences LLM capabilities and computational demands. Increased parameters typically improve performance but require greater computational resources. The proposed framework categorizes scalability into four tiers:

1. *Small-scale models (up to 1 billion parameters)* These models, such as DistilBERT [143], are designed for computational efficiency and deployment in resource-constrained environments. Their smaller size typically enables efficient inference, with reduced latency suitable for edge devices or mobile applications.
2. *Medium-scale models (1 billion to 100 billion parameters)* Models like BERT [29] and GPT-2 [144] balance performance and resource requirements. They provide

strong performance across standard NLP benchmarks, including General Language Understanding Evaluation (GLUE) with BERT achieving an average accuracy of approximately 82% and GPT-2 excelling in generative tasks.

3. *Large-scale models (100 billion to 1 trillion parameters)* This category includes models such as GPT-3 [8] and PaLM [34]. Their large parameter counts enable superior few-shot and zero-shot performance, exemplified by GPT-3 achieving state-of-the-art results on SuperGLUE (approximately 71.8%) and competitive results on MMLU benchmarks. However, they require extensive computational infrastructure, limiting widespread adoption.
4. *Ultra-scale models (1 trillion parameters and above)* These models represent the forefront of LLM research. GPT-4, as a representative model, pushes boundaries in multi-task generalization and reasoning tasks but entails substantial resource consumption, presenting environmental and practical deployment challenges.

8.2 Application domains

Application domain categorization identifies models based on their targeted usage areas:

1. *General-purpose models* These models, like GPT-3 [8], support a wide range of tasks without domain-specific fine-tuning. Their flexibility is beneficial for general language understanding, summarization, and dialogue systems.
2. *Domain-specific models* Specialized for particular fields, models such as BioBERT [145] and LegalBERT [146] offer enhanced accuracy within their domains due to targeted training and fine-tuning. For instance, BioBERT achieves significantly better performance on biomedical question-answering datasets compared to general models.
3. *Task-specific models* Designed explicitly for single tasks, these include Codex [101], which focuses on code generation. Codex demonstrates impressive performance, achieving approximately 72% accuracy on HumanEval, a benchmark for functional code generation.
4. *Multimodal models* Incorporating multiple data types beyond text, multimodal models such as OpenAI's DALL E [147] integrate text and visual modalities. They enable novel applications in text-to-image synthesis, significantly extending the practical utility of LLMs.

8.3 Ethical risk tiers

Lastly, the ethical risks associated with LLMs vary depending on their design, data sources, and applications. This framework introduces ethical risk tiers to assess the potential societal impact of these models:

1. *Low-risk models* Models used in low-stakes environments, such as entertainment chatbots, would fall into this category, as the consequences of bias or misinformation are minimal.
2. *Medium-risk models* This category includes models deployed in controlled environments with moderate ethical concerns, such as educational tools or content recommendation systems.
3. *High-risk models* Deployed in sensitive domains (e.g., healthcare, legal systems), these models demand rigorous audits to ensure fairness, transparency, and accountability.

Biases or inaccuracies in these models can directly affect critical decision-making processes, necessitating stringent regulatory oversight.

4. *Critical-risk models* Utilized in high-stakes environments such as law enforcement or autonomous systems, these models bear significant ethical implications. Continuous monitoring and regulatory governance are essential due to potential severe consequences from misuse or failure.

8.4 Multilingual and cross-lingual capabilities

As the deployment of LLMs expands globally, the ability to perform across multiple languages and generalize to cross-lingual contexts has become a defining feature distinguishing modern LLMs from earlier NLP models. Multilingual and cross-lingual capabilities ensure that language technologies can support inclusive access, knowledge dissemination, and equitable AI deployment across diverse linguistic communities.

Early models, such as word2vec and GloVe, were language-specific, lacking any capacity for multilingual reasoning. The introduction of multilingual BERT (mBERT) marked a turning point by enabling a single transformer model to serve over 100 languages without explicit supervision. A study by Plank and Agić [148] demonstrated that mBERT could transfer syntactic knowledge effectively across English, French, and Italian, showing potential for shared representations across linguistic families. Later efforts, such as XLM [149], introduced cross-lingual pretraining objectives like translation language modeling (TLM) to improve cross-language alignment. XLM-R [150] built upon this with robust pretraining on 2.5TB of CommonCrawl data across 100 languages, significantly outperforming previous models on cross-lingual benchmarks, such as XNLI and MLQA. Similarly, mT5 [151] extended the T5 framework to over 100 languages, incorporating multilingual span corruption as its training objective.

More recent LLMs such as BLOOM [152] and GPT-4 have demonstrated emergent multilingual capabilities, even when not explicitly trained for multilingualism. Gemini 1.5 and Mistral models have also showcased improved performance in low-resource and code-switched scenarios. These models rely on strategies such as shared tokenization schemes (e.g., SentencePiece) and semantic alignment in multilingual embedding spaces. Wang et al. [153] show that aligning multilingual embeddings during training significantly improves in-context learning in low-resource languages. Zhao et al. [154] further highlight that performance in morphologically rich and low-resource languages remains inconsistent in state-of-the-art models, such as ChatGPT, raising concerns around linguistic fairness and representation.

To support evaluation, several multilingual benchmarks have been introduced, including XGLUE [155], XTREME [156], XTREME-R [157], and FLORES-101 [158]. These benchmarks help quantify both in-language and cross-lingual transfer capabilities.

In this taxonomy, multilingual and cross-lingual capability is introduced as a fourth axis of classification, acknowledging the increasing importance of linguistic diversity in AI systems. Models can be categorized based on:

1. *Monolingual models* Trained and optimized for a single language.
2. *Multilingual models* Trained across multiple languages with shared vocabulary and representations (e.g., mBERT, XLM-R, mT5).

Table 2 Summary of the proposed taxonomy for categorizing LLMs with examples

Dimension	Categories	Key characteristics	Examples
Scalability tiers	Small-scale models (Up to 1 B parameters)	Lightweight models suitable for edge devices and low-complexity tasks.	DistilBERT, Alpaca 7B
	Medium-scale models (1–100 B parameters)	Balanced performance and resource consumption, suitable for a wide range of tasks.	BERT, GPT-2
	Large-scale models (100 B–1 T parameters)	Capable of advanced capabilities, such as few-shot learning; require significant resources.	GPT-3, PaLM, Falcon 180B
	Ultra-scale models (1 T+ parameters)	Cutting-edge models with state-of-the-art performance but significant energy/resource demands.	GPT-4, Llama 3.1
Application domains	General-Purpose Models	Designed for broad applicability without domain-specific fine-tuning.	GPT-3, GPT-4
	Domain-specific models	Optimized for specific domains, such as healthcare or legal analysis.	BioBERT, LegalBERT
	Task-Specific Models	Focused on a single task such as search, code generation, or instruction tuning.	Codex, Command R
	Multimodal models	Incorporate modalities beyond text, such as images, audio, or video.	DALL E, Imagen
Ethical risk tiers	Low-risk models	Deployed in non-critical environments where errors have minimal consequences.	Stable LM 2
	Medium-risk models	Used in moderately critical domains such as education or content recommendations.	Falcon 180B, Jamba
	High-risk models	Operate in high-stakes domains, including healthcare or public policy, requiring rigorous audits.	GPT-4, PaLM 2
	Critical-risk models	Used in sensitive applications such as law enforcement or autonomous systems, requiring continuous oversight.	Nemotron-4
Multilingual capabilities	Monolingual models	Trained and optimized for a single language, often achieving high performance within a narrow linguistic context.	BERT (English), CamemBERT (French), GBERT (German)
	Multilingual models	Trained on multiple languages simultaneously with shared vocabulary and representations.	mBERT, XLM-R, mT5
	Cross-lingual models	Designed to support transfer learning across languages; evaluated on zero-shot or few-shot cross-lingual tasks.	XLM, BLOOM, GPT-4

3. *Cross-lingual models* Explicitly optimized for transfer between languages, often evaluated using translation and zero-shot generalization (e.g., XLM, BLOOM, GPT-4).

As multilingual deployment becomes a cornerstone for global AI equity, evaluating and comparing models based on their multilingual performance is vital for both researchers and policymakers.

This structured taxonomy, summarized in Table 2, offers a comprehensive framework for evaluating LLMs by categorizing them along key dimensions that capture both technical specifications and deployment context. The scalability tiers differentiate models based on their parameter size and computational footprint, ranging from lightweight models like DistilBERT and Alpaca 7B, which are suitable for edge devices and low-complexity tasks, to ultra-scale models such as GPT-4 and Llama 3.1, which deliver state-of-the-art performance but require significant infrastructure and energy resources. The application domain dimension supports alignment between model capabilities and use cases, distinguishing general-purpose models like GPT-3 and GPT-4, domain-specific models such as BioBERT and LegalBERT, and task-specific systems, such as Codex and

Command R, with additional consideration for multimodal models including DALL E and Imagen that integrate text with images or other modalities.

Ethical considerations are captured in the risk tiers dimension, which stratifies models by their potential impact and required oversight. For example, Stable LM 2 may be deployed in low-risk settings, whereas models, such as Falcon 180B or Jamba fall into the medium-risk category due to their broader use in education or content recommendation. High-risk and critical-risk models, such as GPT-4, PaLM 2, or Nemotron-4, are used in domains like healthcare, public policy, or autonomous decision-making and thus demand more rigorous governance and ongoing monitoring. Lastly, the multilingual capabilities dimension captures linguistic coverage, distinguishing monolingual models like CamemBERT (French) or GBERT (German) from multilingual systems, such as mBERT, XLM-R, or mT5, and cross-lingual models like XLM, BLOOM, and GPT-4, which support transfer learning and multilingual generalization. Together, these dimensions and representative models provide a structured and practical taxonomy for supporting decisions around model selection, deployment suitability, and regulatory compliance across diverse contexts.

9 Challenges and future research directions

Despite the remarkable advancements achieved by LLMs, they still face challenges that must be addressed to ensure their safe, ethical, and efficient deployment across industries. This section discusses the critical limitations affecting LLMs and outlines future research directions aimed at overcoming these challenges.

9.1 Ethical concerns and mitigating bias

A pressing ethical challenge for LLMs is mitigating bias embedded in their training data. Since these models are trained on extensive datasets scraped from the internet, they inherently inherit societal biases related to race, gender, and other social factors. This can result in outputs that perpetuate harmful stereotypes or exhibit unfair treatment in contexts, such as job recruitment or legal decisions [159]. Addressing these biases requires refining training datasets, developing robust debiasing techniques, and increasing transparency in how LLMs generate outputs.

Future research should focus on designing dynamic debiasing algorithms that adapt during deployment to new contexts, as well as explainability tools that improve user trust by offering clear insights into model decisions [160, 161]. RLHF has shown promise in aligning LLM outputs with human ethical values and societal norms [162], but further exploration is needed to generalize these methods across diverse domains.

9.2 Technical challenges and innovations

The “black box” nature of LLMs poses significant interpretability challenges, particularly in critical fields such as healthcare and law, where decisions must be trustworthy. While techniques such as attention maps and saliency visualizations offer partial insights, they fail to provide a comprehensive explanation of how these models process information [91]. Future work should prioritize developing advanced interpretability frameworks that can elucidate the decision-making process of LLMs, making them more transparent and accountable in sensitive applications.

Another major limitation is the high computational cost of training and deploying LLMs, which creates barriers for smaller organizations and raises environmental concerns. Although methods such as model distillation, pruning, and quantization have been proposed to reduce computational overhead [163], further innovation is required to enhance their scalability and efficiency. Research should also explore architectures that minimize resource consumption, such as sparse models or energy-efficient training algorithms that optimize hardware utilization.

9.3 Deployment risks and hallucination mitigation

LLMs frequently generate factually incorrect but plausible-sounding information, a phenomenon known as "hallucination." This poses significant risks in applications such as automated news generation, legal document drafting, and medical advice, where accuracy is paramount. Future efforts should focus on developing robust content validation mechanisms, including integrating LLMs with external knowledge bases or fact-checking pipelines to cross-reference outputs with reliable sources [164].

Additionally, LLMs struggle with tasks requiring complex reasoning or multi-step problem-solving, which limits their utility in areas such as mathematics, logical puzzles, or abstract reasoning [165]. Future research should explore hybrid architectures that integrate explicit reasoning modules or external symbolic systems, enabling LLMs to handle tasks that demand structured thinking and logical consistency.

9.4 Multimodal integration and cross-domain adaptability

While LLMs have excelled in text-based tasks, their ability to integrate and reason across multiple data modalities, such as images, audio, and video, remains limited. Current systems, such as OpenAI's GPT-4 and DALL E, demonstrate early capabilities in text-to-image generation but lack seamless cross-modal reasoning [166, 167].

Future research should focus on developing unified multimodal models that can process and integrate diverse data types while reasoning cohesively across them. Enhancing cross-domain adaptability will allow models to transition smoothly between tasks, such as combining natural language processing with visual recognition, thereby unlocking new possibilities for practical AI systems.

9.5 Sustainability and green AI

The environmental impact of LLMs has become a growing concern, especially GPT-4 consuming immense computational resources. Techniques such as federated learning and distributed training offer promising paths to reduce the carbon footprint of large-scale AI systems by decentralizing computation and enhancing data privacy [168].

Future work should prioritize developing benchmarks to evaluate the environmental impact of AI systems during training and deployment, incentivizing the creation of lightweight architectures for resource-constrained environments. Emphasis on "green AI" practices, such as energy-efficient parallelism and adaptive training algorithms, can help align the growth of LLMs with sustainable development goals [169, 170].

10 Conclusion

LLMs have transformed the field of natural language processing by enabling a wide range of capabilities–from text generation and translation to domain-specific tasks in law, healthcare, education, and cybersecurity. This review examined foundational architectures such as GPT, BERT, PaLM, LLaMA, and the more recent DeepSeek-R1, demonstrating how advances in transformer-based designs and training methodologies have enabled these models to generalize across diverse tasks with minimal supervision.

Building on these insights, this study introduced a conceptual taxonomy that classifies LLMs based on three interrelated dimensions: scalability, application domains, and ethical risk. This framework offers a structured way to compare models and informs both developers and policymakers on appropriate deployment contexts. The review also outlined limitations in existing literature, especially around underexplored domains such as prompt engineering, scalability trade-offs, and real-world ethical implications.

Furthermore, future research must focus on improving model interpretability, reducing environmental impact, and ensuring alignment with ethical and societal values. Addressing challenges such as dataset contamination, hallucination, and responsible deployment will require both technical innovations and interdisciplinary collaboration involving ethicists, regulators, and domain experts. Lastly, as LLMs continue to evolve, maintaining transparency, accountability, and inclusivity will be essential to their safe and equitable integration into society.

Author contributions

I.D.M. conceptualized the study, designed the methodology, and contributed to the manuscript writing. N.J. and O.O.O., E.E., and C.M. conducted the literature review, performed data analysis, and contributed to manuscript editing. G.O. supervised the research and provided critical revisions. All authors reviewed and approved the final manuscript.

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Declarations

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No consent for participation is required, as no personal data included.

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Competing interests

The authors declare no conflict of interest.

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